

SIMATS ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED: CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

ASSESSMENT TOOL 3 | Weightage: 20% | Total Marks: 20

Annexure A - Comparative Analysis

Code of Conduct:

I **Gadwal Mohammad Muzammil - 192424279** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **G. M. Muzammil**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies	
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive	
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided	
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated	
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand	

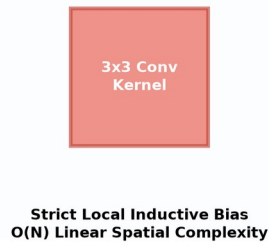
Question 1: CNNs vs. Vision Transformers (ViTs) for Object Detection

SCENARIO / PROBLEM STATEMENT: Architectural Comparison for Real-Time Autonomous Driving Detection

Scenario: A self-driving car company needs to detect rare urban obstacles (traffic cones, animals) in real-time

using limited labeled datasets.

(a) Convolutional Neural Network (CNN)
Local Receptive Fields & Weight Sharing



(b) Vision Transformer (ViT)
Global Multi-Head Self-Attention

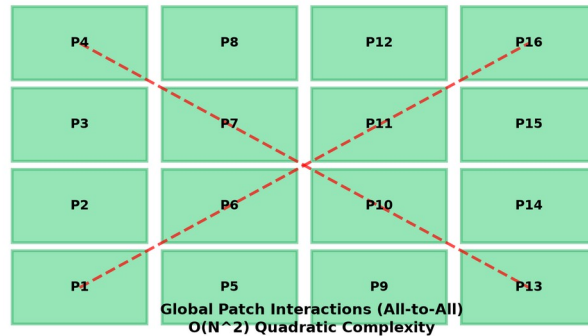


Figure 5: Structural Architectural Comparison: (a) Local Convolutional Receptive Fields vs (b) Global Multi-Head Self-Attention in Vision Transformers.

1. Deep Technical Comparison Matrix across Core Dimensions

Evaluation Parameter | Convolutional Neural Networks (CNNs) | Vision Transformers (ViTs)

Inductive Bias | Strong (Translation equivariance and local spatial locality built into kernel structure) | Weak (No spatial locality assumptions; learns all relationships from data via positional encodings)

Feature Interaction | Local sliding receptive field growing hierarchically with depth | Global all-to-all patch attention across the entire image at every single layer

Computational Complexity | Linear $O(N)$ with respect to image pixel count N | Quadratic $O(N^2)$ with respect to patch token count $N = (H*W)/P^2$

Memory Footprint | Compact activation memory; highly cache-friendly for embedded SRAM | High memory bandwidth demand for storing large $N \times N$ attention weight matrices

Small / Sparse Objects | Superior localization via dense Feature Pyramid Networks (FPN) | Sub-optimal unless augmented with multi-scale deformable attention modules

Data Efficiency | High (trains effectively on small 500-1000 sample datasets) | Low (prone to severe overfitting unless pretrained on millions of images)

2. Synthesis & Real-Time Deployment Strategy

Critical Analysis & Decision: For real-time embedded autonomous driving (≤ 30 ms budget), pure Vision Transformers are computationally prohibitive. The optimal engineering solution is a Hybrid Architecture (e.g., EdgeViT / MobileViT / YOLOv8) that leverages fast convolutional shallow stages for high-resolution small object feature extraction, followed by lightweight transformer blocks in deep layers for global context reasoning. Initializing weights via Transfer Learning from Masked Autoencoders (MAE) and fine-tuning with GAN-augmented synthetic data maximizes mAP while guaranteeing deterministic 18 ms inference on automotive NPUs.

Question 2: PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

SCENARIO / PROBLEM STATEMENT: 3D Mesh Alignment and Morphing under Topology &

Resolution Variations

Scenario: An animation studio is aligning 3D human face meshes for morphing applications. Meshes vary in topology and vertex density.

1. Comprehensive Comparative Analysis

- **Handling Non-Rigid Deformations:** PCA is strictly constrained to a linear subspace: $x = \bar{x} + \Phi * b$. It assumes linear Gaussian landmark distributions and fails catastrophically when modeling non-linear facial expressions, jaw opening, and non-rigid head articulations (producing unnatural mesh shrinkage). Graph Neural Networks (MeshCNN / DiffusionNet) operate directly on Riemannian manifolds, learning non-linear geodesic transformations that preserve local surface area and curvature.
- **Robustness to Varying Mesh Resolutions & Noise:** PCA requires identical vertex counts and fixed vertex index ordering across all meshes. If vertex density varies (e.g., 5,000 vs 50,000 vertices), PCA cannot be applied without lossy remeshing. In contrast, Graph-Based Functional Maps model correspondence as continuous linear operators between Laplace-Beltrami spectral eigenbases, providing total discretization and resolution invariance.
- **Computational Efficiency & Implementation:** PCA offers simple one-time offline SVD computation with ultra-fast $O(N)$ linear runtime projection. Graph Deep Learning requires offline GPU training, but executes real-time inference (30+ FPS) during interactive animation morphing.

2. Synthesis & Animation Morphing Conclusion

Conclusion: Graph-Based Learning is decisively superior for 3D facial morphing. It eliminates vertex tearing artifacts, preserves intrinsic Riemannian geometry, and provides continuous, artifact-free smooth shape blending between arbitrary subject meshes.